

Ordinary Differential Equations Note

Differential Equations with Boundary-Value Problems, 9/e (Metric Edition)

0 Introduction

Example. If $y = f(x)$, then we have the derivative of f is

$$y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{df(x)}{dx}.$$

If the equation is given

$$y' - \cos x = 0,$$

then $y = \sin x$ is a solution.

In particular, $y = \sin x + C$ is also a solution of the equation for any constant C .

Example. If given a equation

$$y'' - 3y' + 2y = 0,$$

then $y = 0$ is a solution, we call it is **trivial solution**.

Also the another equation(RLC)

$$C \frac{d^2 v(t)}{dt^2} + \frac{1}{R} \frac{dv(t)}{dt} + \frac{1}{L} v(t) = \frac{dI_C(t)}{dt},$$

can you find another solutions?

1 Introduction to Differential Equations

1.1 Definitions and Terminology

Definition. The differential equations have two forms,

1. **Ordinary differential equations** are the form with only one variable. For example,

$$ay''(x) + by'(x) + cy(x) = g(x)$$

is an ordinary differential equation.

2. Two or more variables, we call it is **partial differential equation**.

$$\alpha \frac{\partial^2 f}{\partial x^2} + \beta \frac{\partial^2 f}{\partial x \partial y} + \gamma \frac{\partial f}{\partial y} = \zeta(x, y)$$

is a partial differential equation.

Definition.

1. The **order** of differential equation is defined as the highest order of derivative.
2. The **degree** of differential equation is defined as the degree of the highest order of derivative.

Example. What is the order and degree of the equation?

$$y'' + 3x^2 = 2(y')^2$$

Sol. The highest order of derivative is y'' , and its degree is 1st degree. Hence the equation is **second order, first degree ODE**.

Definition.

1. The equation is **linear** if the dependent variable (y or others) is only first degree.

$$a_n y^{(n)} + \cdots + a_1 y' + a_0 y = g(x).$$

2. The equation is **nonlinear** if it is not linear equation.

Definition.

1. A **homogeneous equation** is the equation where all terms involving the dependent variable equal zero.

$$a_n y^{(n)} + \cdots + a_2 y'' + a_1 y' + a_0 y = 0.$$

2. A **nonhomogeneous equation** is the equation that it is not homogeneous equation.

Example. Determine the differential equations.

(a) $x^2 y'' + \sin y \cdot y' + y^2 = 60e^x$

(b) $y' + \frac{y}{x} = xy^2$

(c) $y' = \frac{x+y}{x-y}$

Sol.

- (a) Second order, first degree, nonlinear and nonhomogeneous ODE.
- (b) First order, first degree, nonlinear and nonhomogeneous ODE.
- (c) First order, first degree, nonlinear and nonhomogeneous ODE.

Example. Determine the differential equations.

(a) $(3x^2 + y \cos x)dx + (\sin x)dy = 0$

(b) $d(yu) = y^2 du$

Sol.

- (a) First order, first degree, linear and homogeneous ODE.
- (b) Since it is

$$\frac{d}{du} yu = \frac{dy}{du} + y = y^2,$$

Hence it is first order, first degree, nonlinear and homogeneous ODE.

1.2 Initial-Value Problems

Example. The equation

$$y' = \cos x$$

has at least one solution.

$$y = \sin x$$

is a solution of the equation.

$$y = \sin x + C$$

is also a solution of the equation, for C is constant.

1. From the example, we call

$$y = \sin x + C$$

is **general solution**.

2. If there is an initial condition on the equation, then the solution is called **particular solution**, for instance,

$$y = \sin x + 1, \quad \text{if } y(0) = 1.$$

3. Singular solution is:

- (1) A solution of the ODE with no constant.
- (2) Cannot derived from the general solution by assigning a constant.
- (3) [Example](#)

Definition. (Expression form of solution)

1. **Explicit form** is shown as

$$y = f(x).$$

2. **Implicit form** is shown as

$$g(x, y) = c.$$

3. **Parametric form** is shown as

$$x = x(t), y = y(t).$$

Definition.

1. The equation with initial condition(s), we say it is **initial value problems**(IVP).
2. **Boundary value problem**(BVP)¹ is an equation with some boundary conditions, they can be some points or intervals.

¹與初值問題 (Initial Value Problem, IVP) 的差異: IVP 的所有條件都給定在同一個起始點 (例如 $t = 0$) ; 而 BVP 的條件則分散在自變數的各個不同邊界點上 (例如 $x = 0$ 與 $x = L$) 。

Example.

1. The equation

$$y = \sin x + 1, y(0) = 1$$

is an initial value problems.

2. The equation

$$y'' + 2y = 0, y(0) = 0, y(10) = 1$$

is a boundary value problem.

1.2.1 The existence and uniqueness of solutions

Definition. The **interval of definition of a solution** is an interval for solution of equation that satisfies the domain of function.

Example. The equation

$$y' + 2xy^2 = 0$$

with $y(0) = -1$.

Sol. The general solution of equation is

$$y = \frac{1}{x^2 + C}.$$

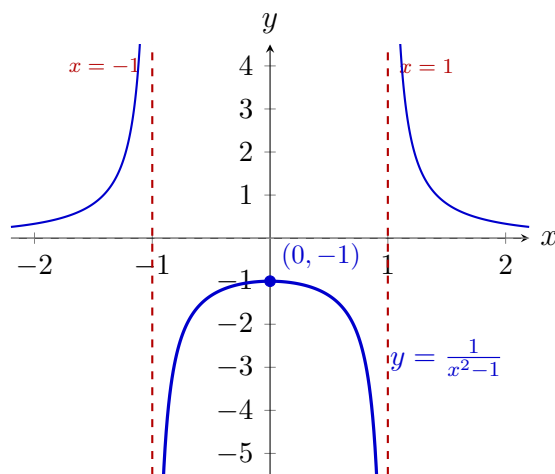
Let $y(0) = -1$, we obtain

$$y = \frac{1}{x^2 - 1},$$

thus $y = \frac{1}{x^2 - 1}$ is a particular solution of the equation.

- (1) The domain of $y = \frac{1}{x^2 - 1}$ is $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$.

- (2) It is clear that $0 \in (-1, 1)$. Excluding $(-\infty, -1)$ and $(1, \infty)$, $(-1, 1)$ is the interval of definition of solution.



Remark. An IVP can have not only one solution. $y' = x\sqrt{y}$ with $y(0) = 0$ is the unique solution.

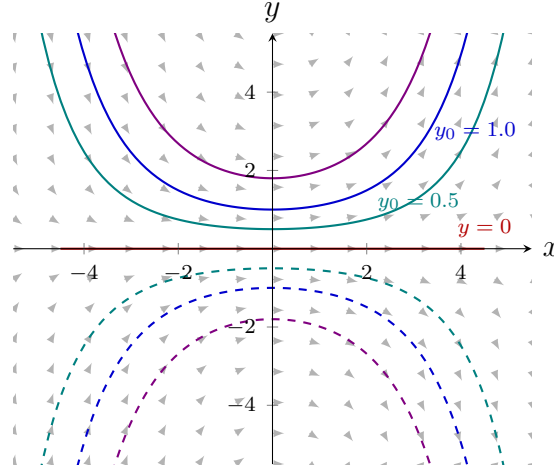
1.2.2 Direction Field(方向場)

Find solutions with the direction field is a qualitative analysis(定性分析).

Example. The equation

$$y' = 0.2xy.$$

With the direction field and solution curve:



1.2.3 Phase portrait for Autonomous ODE(自主性方程與相圖)

To solve the equation shown as the form

$$\frac{dy}{dx} = f(y),$$

we will use the following steps:

1. Equilibrium(平衡) / Critical Points: Let $y' = f(y) = 0$ and find its critical points.

Example.

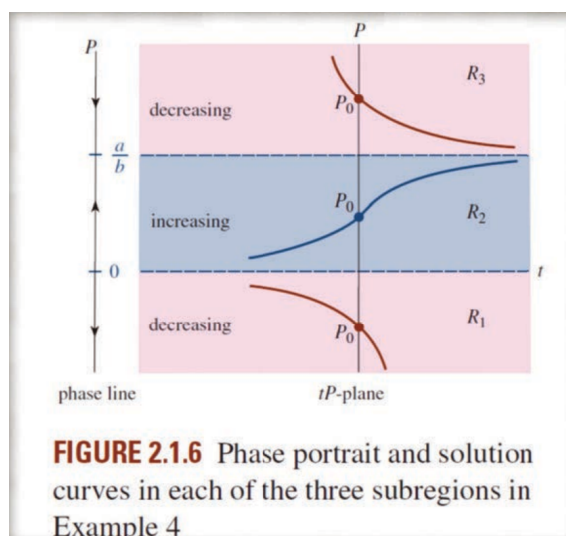
$$\frac{dp}{dt} = p(a - bp) = f(p).$$

Let $p' = p(a - bp) = f(p) = 0$, implies

$$p(a - bp) = 0.$$

Then $p = 0$ or $p = \frac{a}{b}$.

2. Phase portrait



The **attractor**(吸引子) is a point that both arrows on either side of the phase line point to this point, which is the stable equilibrium point (e.g. $p = \frac{a}{b}$); and the **repeller**(排斥子) is a point that the arrows on both sides of the phase line are moving away from the point, indicating an unstable equilibrium point. (e.g. $p = 0$)